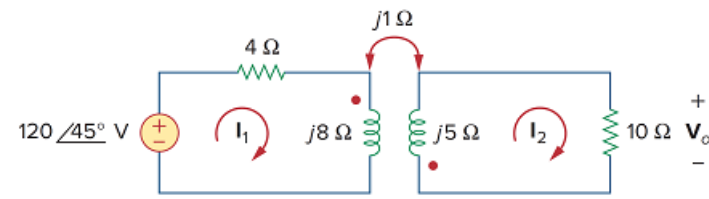


Problem

Determine the voltage V_o in the circuit of Fig. 13.10.

Figure 13.10



Step-by-step solution

Step 1 of 3

Refer Figure 13.10 in the textbook for the actual circuit.

Redraw circuit diagram as shown in Figure 1.

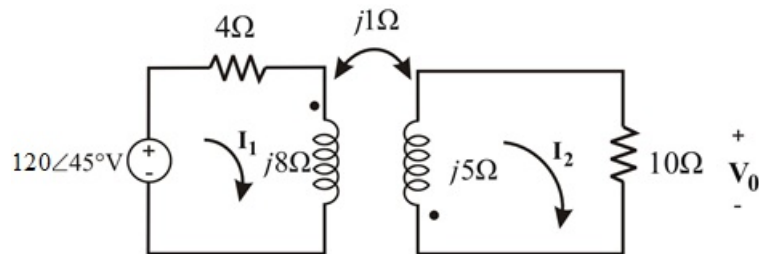


Figure 1

Step 2 of 3

Clearly from Figure 1, both currents I_1, I_2 are entering at dotted terminal.

Applying KVL to first loop:

$$-120\angle 45^\circ + (4 + j8)I_1 + j1I_2 = 0$$

$$(4 + j8)I_1 + j1I_2 = 120\angle 45^\circ \dots\dots(1)$$

Applying KVL to second loop:

$$(10 + j5)I_2 + j1I_1 = 0$$

$$I_1 = -\frac{(10 + j5)}{j}I_2$$

$$I_1 = (-5 + j10)I_2$$

Step 3 of 3

Substitute $(-5 + j10)\mathbf{I}_2$ for $\mathbf{I}_1$ in equation (1).

$$(4 + j8)(-5 + j10)\mathbf{I}_2 + j1\mathbf{I}_2 = 120\angle 45^\circ$$

$$(-100 + j)\mathbf{I}_2 = 120\angle 45^\circ$$

$$\mathbf{I}_2 = \frac{120\angle 45^\circ}{100\angle 179.42^\circ}$$

$$\cong \frac{120\angle 45^\circ}{100\angle 180^\circ}$$

$$\mathbf{I}_2 \cong 1.2\angle -135^\circ \text{ A}$$

Calculate the circuit voltage $\mathbf{V}_o$.

Write the expression for voltage $\mathbf{V}_o$.

$$\mathbf{V}_o = 10\mathbf{I}_2$$

$$= 10(1.2\angle -135^\circ)$$

$$= 12\angle -135^\circ \text{ V}$$

Therefore, the value of the voltage $\mathbf{V}_o$ is $\boxed{12\angle -135^\circ \text{ V}}$.